

Collatz-3

Overview

Collatz-3 is a playful yet mathematically rich variant of the classic Collatz map, redesigned around the arithmetic structure of modulo 3.

Each integer follows a simple transformation rule based on its remainder when divided by 3:

- numbers divisible by 3 fall directly through division,
- numbers congruent to 1 or 2 trigger a $4n \pm 1$ transformation, often causing temporary growth.

Despite its simplicity, the system produces surprisingly diverse behaviors.

Some numbers descend smoothly toward 1, while others wander through long, chaotic trajectories before finally settling.

This tendency to resist the modulo-3 world is measured as **3-Resistance** — the number of steps required to reach 1.

The goal of the Collatz-3 exploration is straightforward:

Identify the integer that resists 3 the most — the Boss.

Within the range 1–100, the champion is **28**, whose trajectory climbs as high as 153 before returning.

The resulting landscape of stopping times reveals valleys, ridges, and isolated peaks, forming a distinctive 3-adic terrain that differs from the classical Collatz map.

Collatz-3 stands as an example of **playable mathematics**: a simple rule that invites experimentation, visualization, and discovery.

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